COMP 6721 Course Project Proposal

Project title: Mutational Signature Detective: Estimating Carcinogen Exposure from Tumor Mutation Patterns

Topic mapping: Custom topic, framed as ML-3/ML-4 (imbalanced, explainable signal decomposition) applied to genomics. Instructor approval requested.

Team:

- Anas Farooqui , 40347862, Contact person
- Hugo Xu (40064100)
- Bao Tran Nguyen, 40257379

---

1. Problem statement

Carcinogens (tobacco, UV, ageing) leave characteristic mutation patterns in tumor DNA, called mutational signatures. Given a tumor's mutation catalog (counts across 96 trinucleotide contexts, the standard SBS representation), we estimate exposure weights over known reference signatures (e.g., SBS4 = tobacco, SBS7 = UV, SBS1/SBS5 = ageing).

We do not claim clinical or legal proof of exposure; some signatures (ageing) aren't external exposures at all. The aim is to estimate and compare attribution methods, and characterize when attribution is reliable.

Research question: How accurately can signature exposure be recovered, at what mutation-count threshold does accuracy break down, and does performance hold on tumors from an unseen cohort?

2. Data and resources

- **Reference signatures:** SBS catalog via SigProfilerMatrixGenerator/SigProfilerExtractor (pip-installable, no registration).
- **TCGA (GDC Data Portal), open-access MAF files, MuTect2-called for consistency.** 5-6 cancer types spanning signature diversity and mutation-burden extremes: lung adenocarcinoma (tobacco), melanoma (UV), colorectal/endometrial (mismatch-repair, high burden), a pediatric or blood cancer (very low burden), plus one more ageing-dominated type.

- **TCGA clinical metadata** (smoking status, subtype) for real-data checks only, not ground truth.
- **PCAWG open-access mutation calls:** independent cohort for a cross-cohort generalization test.
- **SigProfilerAssignment / deconstructSigs:** published methods used for comparison, not reimplemented.

All sources are public, require no access request or license agreement, and no restricted data will be uploaded to Git.

Label caveat: real tumors have no confirmed ground-truth exposure weights; clinical fields are imperfect proxies. Primary validation therefore uses simulation; real data gives directional checks only.

3. Baseline and methods

- **Baseline:** NNLS fitting against the fixed signature matrix (standard classical approach).
- **Compared methods:**
    1. Regularized/constrained fitting (e.g., elastic-net NNLS), for low-mutation robustness.
    2. Probabilistic/Bayesian fitting, giving uncertainty over exposure weights.
    3. A small neural network trained on simulated tumors, as a core method (not a stretch goal), expected to make different errors than the NNLS family.
- **Analysis:** a signature-by-signature confusion matrix from simulation (true vs. attributed), and how confusion changes with mutation count.

4. Evaluation plan

- **Primary (simulation):** synthetic tumors with known signature mixtures, calibrated to real TCGA mixture and mutation-burden statistics rather than sampled arbitrarily. Metrics: cosine similarity and mean absolute error vs. ground truth. Key result: accuracy vs. mutation count, plus the confusion matrix.
- **Secondary (real data, statistical test):** in TCGA lung adenocarcinoma, test whether estimated tobacco-signature weight differs significantly by smoking status (Mann-Whitney U), as a falsifiable directional check, not an accuracy claim.
- **Cross-cohort test:** fit/train on TCGA, evaluate on PCAWG (and reverse if time allows), to detect reliance on cohort-specific artifacts.
- **Sanity check:** benchmark our NNLS baseline against SigProfilerAssignment/deconstructSigs on simulated data before comparing stronger methods.

5. Division of work

- **Member 1:** Data acquisition/preprocessing (TCGA + PCAWG MAF parsing, signature matrix setup, context extraction, caller consistency).
- **Member 2:** Simulation framework; classical/probabilistic methods (NNLS, regularized, Bayesian).
- **Member 3:** Neural network method; evaluation pipeline (confusion matrix, cross-cohort test, statistical test).
- All members share report, reproducibility package, and presentation work.

6. Plan

Weeks 3-8: Install tools; download TCGA (MuTect2) and PCAWG data for selected cancer types; build context-extraction pipeline; build realistic simulator; implement and validate NNLS baseline; first accuracy-vs.-mutation-count results; private Git repo with README from week one.

Weeks 8-13: Implement regularized, Bayesian, and neural-network methods; run confusion-matrix and mutation-count sensitivity analyses; run the smoking-status statistical test and TCGA-PCAWG cross-cohort test; write report, finalize reproducibility package, prepare presentation.

7. Risks and fallbacks

1. **Overreaching exposure claims.** Mitigation: report outputs as estimated contributions, not diagnoses; note ageing isn't an exposure.
2. **No real ground truth.** Mitigation: simulation is the primary validation; real data is explicitly qualitative.
3. **Low-mutation tumors are hard cases.** Mitigation: treat this as an analyzed, expected result, not a bug.
4. **Domain learning curve (genomics, MAF format).** Mitigation: budget weeks 1-2 for this, using SigProfiler/deconstructSigs docs as references.
5. **Data/package friction.** Mitigation: no registration required anywhere; swap in a better-documented cohort if a specific cancer type's files are hard to parse.
6. **Scope risk (6 cancer types + PCAWG + neural net) may not all fit.** Mitigation: the guaranteed minimum is 3 methods on 2-3 cancer types with simulation validation; extra cancer types, PCAWG, and the statistical test are staged extensions, cut first if behind schedule.

8. Limitations

- Signatures indicate mutational processes, not confirmed exposures; ageing isn't external.
- Real-data validation is indirect (clinical proxies), not ground truth.

- Attribution is known to be unreliable at low mutation counts; we characterize rather than hide this.
- This is a research prototype for hypothesis generation, not a diagnostic tool.

9. Ethics and integrity

Only open-access public data is used; no controlled-access data is accessed or uploaded. External tools and papers (SigProfilerMatrixGenerator/Extractor, Alexandrov et al. 2020, TCGA/GDC, PCAWG, SigProfilerAssignment, deconstructSigs) will be cited.